

PROJECT BASED LEARNING

DATABASE MANAGEMENT SYSTEM



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School of Technology

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HOSTEL ACCOMMODATION AND STUDENT SERVICES MANAGEMENT SYSTEM

PAGE 1 – COVER PAGE

HOSTEL ACCOMMODATION AND STUDENT SERVICES MANAGEMENT SYSTEM

A Database Management System Project Report

Submitted by:

Kartikeya Parcha

Program: B.Tech Artificial Intelligence and Machine Learning

Section: AIML Panthers

University: Woxsen University – School of Technology

Academic Year: 2026

Technologies Used:

- MySQL
 - Python
 - Flask
 - HTML
 - CSS
-

PAGE 2 – CERTIFICATE

CERTIFICATE

This is to certify that **Kartikeya Parcha**, a student of **B.Tech Artificial Intelligence and Machine Learning, AIML Panthers, Woxsen University – School of Technology**, has

successfully completed the project titled:

“Hostel Accommodation and Student Services Management System”

as part of the academic requirements for the Database Management System course.

The project demonstrates the design and implementation of a database-driven hostel management application using **MySQL as the database management system** and **Python Flask as the application framework**.

The project provides facilities for managing students, hostels, rooms, beds, allocations, payments, complaints, maintenance requests, services, service requests, staff, visitors, and vacating records.

The work presented in this project report is carried out by the student as part of the academic curriculum.

Faculty Signature: _____

Date: 6/10/26

Place: Woxsen University

PAGE 3 – DECLARATION

DECLARATION

I, **Kartikeya Parcha**, student of **B.Tech Artificial Intelligence and Machine Learning, AIML Panthers, Woxsen University – School of Technology**, hereby declare that the project entitled:

“Hostel Accommodation and Student Services Management System”

is my academic project developed as part of the Database Management System course.

The project focuses on designing and implementing a database system capable of managing hostel accommodation and student-related services.

The system was developed using **MySQL, Python, Flask, HTML, and CSS**. The database contains separate tables for different hostel operations such as student information, hostels, rooms, beds, allocations, payments, complaints, maintenance requests, services, service requests, staff, visitors, and vacating records.

I declare that the information presented in this report represents the work carried out for this project.

Student Name: Kartikeya Parcha

Date: 9/10/26

PAGE 4 — ACKNOWLEDGEMENT ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Woxsen University – School of Technology** for providing me with the opportunity to develop this Database Management System project.

I would like to thank my faculty members for their guidance and support throughout the development of the **Hostel Accommodation and Student Services Management System**.

I am grateful for the academic resources and laboratory facilities provided by the university, which helped me understand database concepts and apply them to a practical application.

I would also like to thank my classmates and friends for their suggestions and support during the development and testing of the system.

Finally, I would like to thank everyone who contributed directly or indirectly to the successful completion of this project.

Kartikeya Parcha

B.Tech Artificial Intelligence and Machine Learning

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PAGE 5 — ABSTRACT ABSTRACT

The **Hostel Accommodation and Student Services Management System** is a database-driven application developed to manage hostel accommodation and various student services in an organized manner.

Traditional hostel management may involve maintaining student information, room allocation, payments, complaints, maintenance requests, and visitor details manually. Managing these records manually can make it difficult to maintain accurate information and quickly retrieve required records.

The proposed system uses a centralized database to store and manage hostel-related information. The database is implemented using **MySQL**, while the application interface is developed using **Python Flask, HTML, and CSS**.

The system contains multiple modules. Student information is maintained through the Students table. Hostel information is stored in the Hostels table, while room and bed details are managed using Rooms and Beds. Allocations maintain information about which student is assigned to a particular bed.

The system also provides modules for payments, complaints, maintenance requests, services, service requests, staff, visitors, and vacating records.

The application supports three major user roles:

- **Student**
- **Warden**
- **Administrator**

Students can view their accommodation information, make payments, raise complaints, submit maintenance requests, and request services. Wardens can manage students, accommodation, complaints, maintenance requests, and service requests. Administrators can manage and view important student and financial information.

The main objective of the project is to provide a simple, centralized, and efficient system for managing hostel accommodation and student services.

PAGE 6 — INTRODUCTION

1. INTRODUCTION

Hostel management is an important administrative activity in universities and educational institutions. A hostel contains a large amount of information related to students, rooms, beds, payments, complaints, maintenance activities, visitors, and other services.

Managing these records manually can become difficult when the number of students increases. Paper-based records can also make searching, updating, and maintaining information time-consuming.

The **Hostel Accommodation and Student Services Management System** is designed to solve these problems by providing a centralized database-based solution.

The system stores hostel-related information in a structured MySQL database. The Flask application provides an interface through which different users can interact with the database.

1.1 Purpose of the System

The purpose of this project is to develop a system that can:

- Store student information.
- Manage hostel information.
- Manage rooms and beds.
- Allocate beds to students.
- Record hostel payments.
- Manage student complaints.
- Manage maintenance requests.
- Manage hostel services.
- Handle service requests.
- Store staff information.
- Record visitor information.
- Maintain vacating records.

1.2 Need for the System

A computerized hostel management system helps reduce manual work and makes information easier to access.

The system provides centralized storage of information and allows authorized users to perform different operations based on their roles.

1.3 Objectives

The main objectives are:

1. To design a database for hostel management.
2. To maintain student accommodation information.
3. To manage rooms and beds.
4. To provide student service request functionality.
5. To manage complaints and maintenance requests.

6. To maintain payment information.
 7. To provide role-based access.
 8. To improve the organization of hostel records.
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PAGE 7 — PROBLEM STATEMENT AND REQUIREMENTS

2. PROBLEM STATEMENT

Hostel administration involves handling large amounts of student and accommodation information.

Without a centralized database, information such as room allocation, payment records, complaints, maintenance requests, and visitor information can become difficult to manage.

The project therefore aims to develop a database management system that provides a centralized platform for managing hostel accommodation and student services.

2.1 Functional Requirements

The system should provide the following functions:

Student Management

The system should store:

- Student ID
- Name
- Gender
- Date of Birth
- Phone
- Email
- Department
- Year of Study

Hostel Management

Student PortalLogout

Welcome, Rahul Kumar

Student ID: 1

Department

CSE

Year

2

Phone

9876543210

Email

rahul@gmail.com

My Room Allocation

Hostel	Room	Bed	Allocation Date	Status
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Pay Hostel Fee

Amount

Enter amount

Pay Fee

Raise Complaint

Complaint Type

Example: Electricity

Description

Describe your complaint

Hostel Management

The system should store:

- Hostel ID
- Hostel Name
- Location
- Warden Name
- Capacity

Administrator Dashboard

Welcome, admin

Total Students

4

Total Fees Collected

₹3734.00

Total Complaints

1

Service Requests

1

Administrator Functions



View Students



Add Student



View Payments



View Complaints

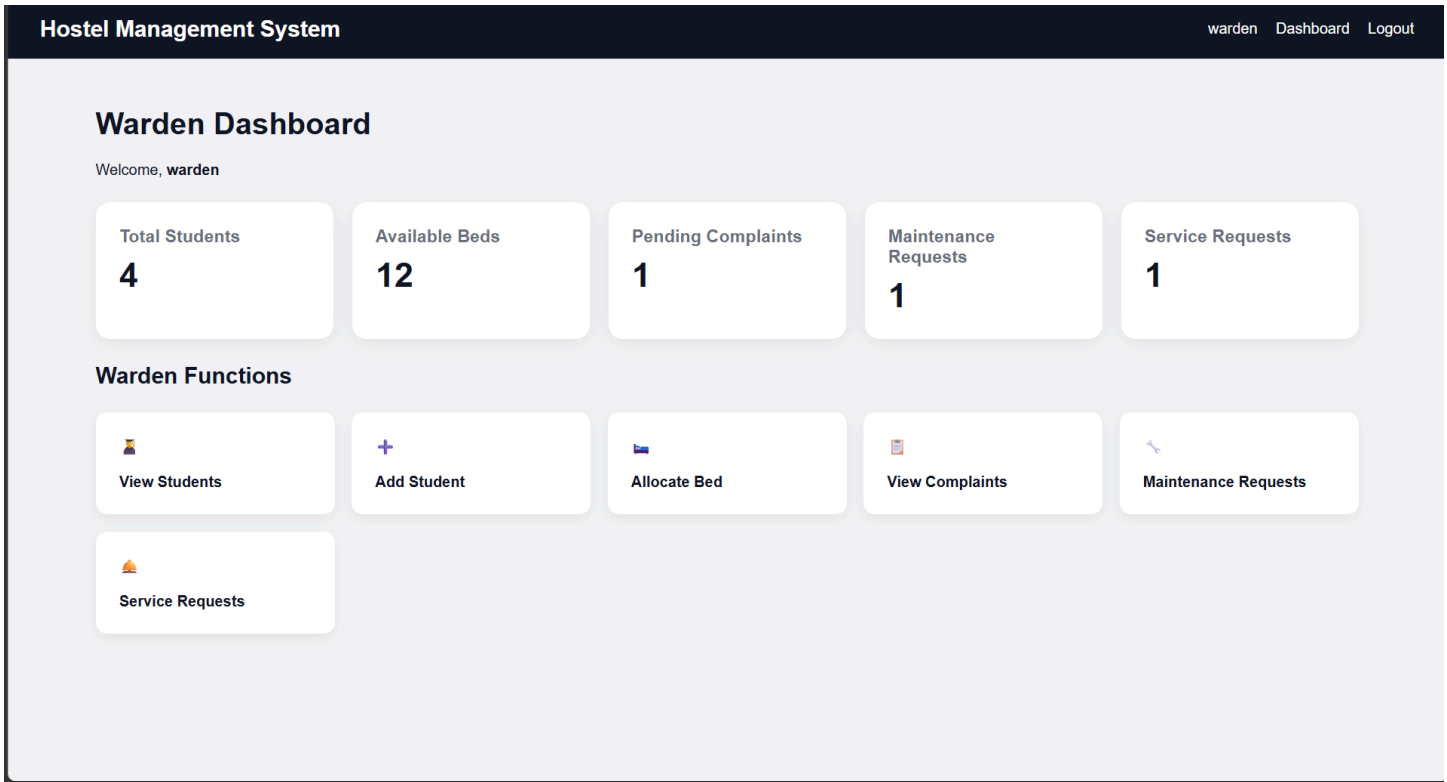


View Service Requests

Room Management

The system should maintain:

- Room ID
- Hostel ID
- Room Number
- Floor
- Room Type
- Capacity



Bed Management

The system should maintain:

- Bed ID
- Room ID
- Bed Number
- Bed Status

Allocation Management

The system should maintain:

- Allocation ID
- Student ID
- Bed ID
- Allocation Date
- Vacate Date
- Allocation Status

Payment Management

Payment records include:

- Payment ID
- Student ID
- Amount

- Payment Date
- Payment Type
- Payment Status

Complaint Management

Complaints contain:

- Complaint ID
- Student ID
- Complaint Type
- Description
- Complaint Date
- Status

Maintenance Management

Maintenance requests contain:

- Request ID
 - Student ID
 - Room ID
 - Issue
 - Description
 - Request Date
 - Status
-

PAGE 8 — SYSTEM ARCHITECTURE

3. SYSTEM ARCHITECTURE

The system follows a database-driven web application architecture.

The major components are:

3.1 User Interface

The user interface is developed using:

- HTML
- CSS

It provides pages for login, dashboards, forms, tables, and management operations.

3.2 Application Layer

The application layer is developed using **Python Flask**.

Flask handles:

- User login
- User sessions
- Routing
- Form submissions
- Database operations
- Role-based access

3.3 Database Layer

The database layer uses **MySQL**.

The MySQL database stores all hostel and student-related records.

3.4 Basic Flow

The system works according to the following flow:

User

↓

Login Page

↓

Role Verification

↓

Student / Warden / Administrator Dashboard

↓

Application Operations

↓

The system uses role-based access so that different users receive different functionality.

3.5 User Roles

Student

Students can:

- View accommodation information
- Make payments
- Raise complaints
- Submit maintenance requests
- Request services
- View their records

Warden

Wardens can:

- View students
- Manage accommodation
- Allocate beds
- Manage complaints
- Manage maintenance requests
- Manage service requests

Administrator

Administrators can:

- View student information
- Manage student details
- View payment information
- View complaints
- View service requests

4. DATABASE DESIGN

The database is named:

HostelDB

The system contains the following tables:

1. Students
2. Hostels
3. Rooms
4. Beds
5. Allocations
6. Payments
7. Complaints
8. MaintenanceRequests
9. Services
10. ServiceRequests
11. Staff
12. Visitors
13. VacatingRecords
14. Users

4.1 Students Table

The Students table stores student information.

Attribute	Description
StudentID	Unique student identifier
Name	Student name
Gender	Student gender
DateOfBirth	Date of birth
Phone	Contact number
Email	Email address
Department	Academic department
YearOfStudy	Current year

4.2 Hostels Table

The Hostels table stores hostel information.

Attribute	Description
HostelID	Unique hostel identifier
HostelName	Name of hostel
Location	Hostel location
WardenName	Warden name
Capacity	Hostel capacity

4.3 Rooms Table

The Rooms table stores room details.

Attribute	Description
RoomID	Unique room identifier
HostelID	Related hostel
RoomNumber	Room number
Floor	Floor number
RoomType	Type of room
Capacity	Room capacity

4.4 Beds Table

The Beds table stores individual bed information.

Attribute	Description
BedID	Unique bed identifier
RoomID	Related room
BedNumber	Bed number
Status	Availability status

PAGE 10 – ENTITY RELATIONSHIPS

5. ENTITY RELATIONSHIPS

The database contains several relationships between the entities.

Student and Allocation

A student can have an allocation record. The Allocations table contains StudentID to associate an allocation with a student.

Hostel and Room

A hostel contains multiple rooms.

The relationship is:

Hostel → Rooms

The HostelID in the Rooms table connects each room with its hostel.

Room and Bed

A room contains multiple beds.

The relationship is:

Room → Beds

The RoomID in the Beds table connects a bed with its room.

Student and Payment

A student can have payment records.

The relationship is:

Student → Payments

StudentID in the Payments table identifies the student associated with a payment.

Student and Complaint

A student can submit complaints.

The relationship is:

Student → Complaints

StudentID connects the complaint to the student.

Student and Maintenance Request

Students can submit maintenance requests related to their rooms.

The MaintenanceRequests table contains StudentID and RoomID.

Student and Service Request

Students can request available hostel services.

The ServiceRequests table contains StudentID and ServiceID.

Student and Visitor

Visitor records are connected to students using StudentID.

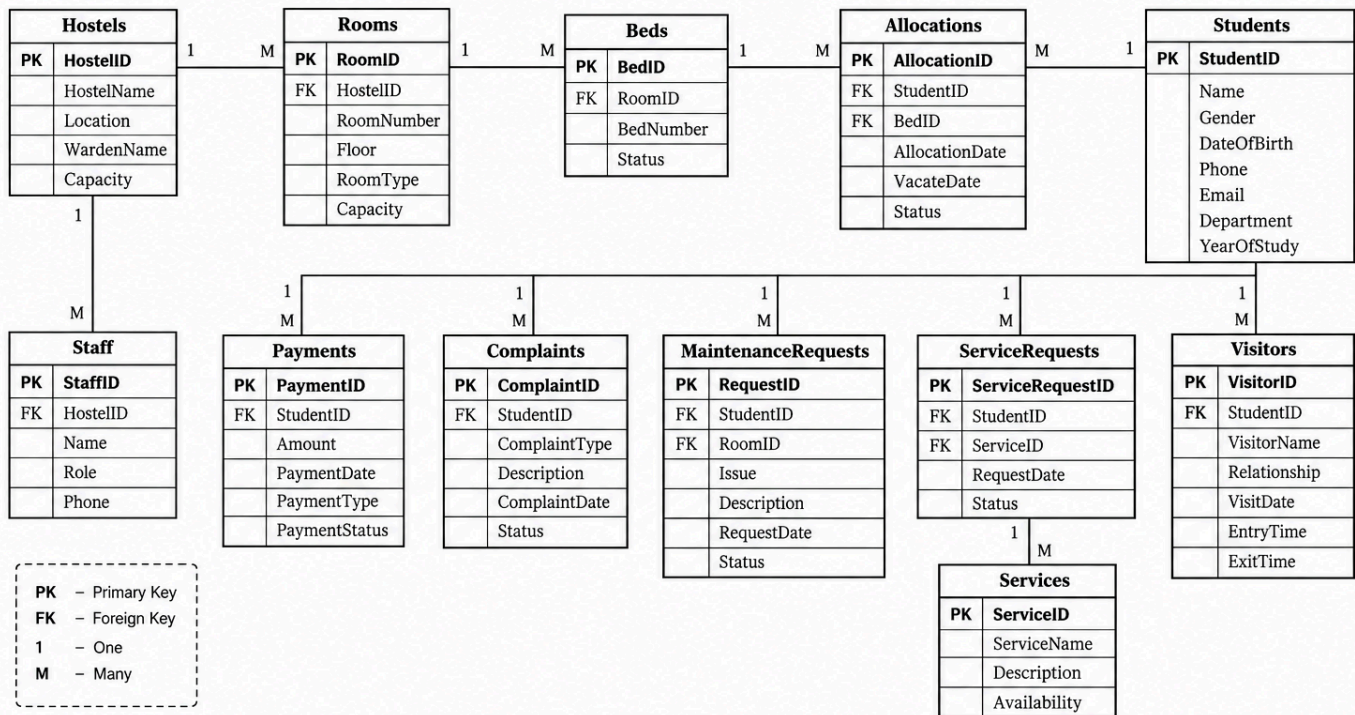
Student and Vacating Record

Vacating records contain StudentID and BedID to record when a student vacates accommodation.

Main Relationship Structure

Database Design

The database is designed using a relational model. It consists of 12 tables that store information about students, hostels, rooms, beds, accommodation, payments, complaints, maintenance requests, services, staff and visitors. The tables are related using primary keys (PK) and foreign keys (FK) to maintain data consistency and reduce redundancy.



PAGE 11 – IMPLEMENTATION

6. IMPLEMENTATION

The system was implemented using **Python Flask and MySQL**.

6.1 Technologies Used

MySQL

MySQL is used as the database management system. It stores all project information in tables.

Python

Python is used for application development and database interaction.

Flask

Flask is used as the web framework. It handles application routes and user requests.

HTML

HTML is used to create the structure of the web pages.

CSS

CSS is used to design the interface and improve the appearance of the application.

6.2 Login System

The system contains a login mechanism using the Users table.

The Users table contains:

- UserID
- Username
- Password
- Role

The three available roles are:

Student

Username: student1

Password: student123

Warden

Username: warden

Password: warden123

Administrator

Username: admin

Password: admin123

After successful login, the user is redirected to the appropriate dashboard.

6.3 Database Connectivity

The Flask application connects to MySQL using the MySQL Connector library.

The application establishes a connection to:

Host: localhost

Database: HostelDB

Database queries are executed from the Flask application to retrieve and modify information.

PAGE 12 — USER MODULES

7. USER MODULES

The application provides different modules according to the user's role.

7.1 Student Module

The Student dashboard provides access to student-related operations.

Students can:

View Accommodation

Students can view their hostel, room, and bed allocation information.

Make Payment

Students can submit payment information.

Raise Complaint

Students can submit complaints by providing the complaint type and description.

Complaint status initially appears as:

Pending

The warden can later update the status.

Maintenance Request

Students can submit maintenance issues related to their allocated room.

Maintenance requests can have statuses such as:

- Pending
- In Progress
- Completed
- Rejected

Service Request

Students can request available hostel services.

Service request statuses include:

- Requested
 - Approved
 - Completed
 - Rejected
-

7.2 Warden Module

The Warden dashboard provides hostel management functionality.

The warden can:

- View students
- Add students
- Allocate beds
- View allocations
- Manage complaints
- Manage maintenance requests
- Manage service requests

Complaint Management

The warden can update complaints from:

Pending → Resolved

or

Pending → Rejected

Maintenance Management

The warden can update maintenance requests:

Pending → In Progress → Completed

The request can also be rejected when necessary.

Service Request Management

Service requests can be updated:

Requested → Approved → Completed

A request can also be rejected.

7.3 Administrator Module

The Administrator dashboard provides access to administrative information.

The administrator can:

- View students
- Add student information
- Edit student information
- View payments
- View complaints
- View service requests

The administrator provides overall administrative control over important records.

PAGE 13 — TESTING AND RESULTS

8. TESTING AND RESULTS

Testing was performed to verify that the main functions of the system work correctly.

Test Case	Expected Result	Result
Student Login	Student dashboard opens	Pass
Warden Login	Warden dashboard opens	Pass
Administrator Login	Admin dashboard opens	Pass
Add Student	Student record added	Pass
View Students	Student records displayed	Pass
Allocate Bed	Allocation recorded	Pass
Student Payment	Payment record created	Pass
Raise Complaint	Complaint created as Pending	Pass
Maintenance Request	Request created	Pass
Service Request	Service request created	Pass
Update Complaint	Status changes	Pass
Update Maintenance	Status changes	Pass
Update Service Request	Status changes	Pass
View Allocation	Allocation information displayed	Pass
Vacating Record	Vacating information stored	Pass

8.1 Complaint Status Testing

A complaint submitted by a student initially has:

Pending

The warden can change it to:

Resolved

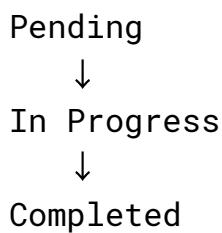
or

Rejected

This allows the student issue to be tracked through the system.

8.2 Maintenance Status Testing

Maintenance requests use the following workflow:

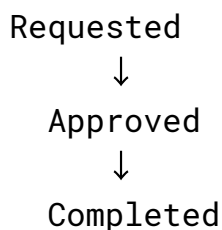


A request may also be:

Pending → Rejected

8.3 Service Request Testing

Service requests use:



A request can also be rejected.

8.4 Result

The testing confirms that the major database operations and application modules work according to the requirements of the project.

PAGE 14 — ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

9. ADVANTAGES

The Hostel Accommodation and Student Services Management System provides several advantages.

9.1 Centralized Data

All hostel-related information is stored in a centralized database.

9.2 Easy Record Management

Student, room, bed, payment, complaint, and service records can be managed more efficiently.

9.3 Faster Information Retrieval

Database queries allow information to be retrieved quickly.

9.4 Role-Based Access

Different users receive different functionality according to their roles.

9.5 Reduced Manual Work

The system reduces dependence on paper-based records.

9.6 Request Tracking

Complaints, maintenance requests, and service requests can be tracked using their status.

9.7 LIMITATIONS

The current system has some limitations.

1. The application is designed primarily for academic demonstration.
 2. The database is hosted locally.
 3. Advanced notification systems are not implemented.
 4. Payment processing is represented through database records rather than an online payment gateway.
 5. The system does not currently provide a dedicated mobile application.
 6. Advanced reporting and analytics can be added in the future.
-

9.8 FUTURE SCOPE

The project can be expanded with additional functionality.

Mobile Application

A mobile application could allow students to access hostel services using smartphones.

Online Payment Gateway

Real payment gateways could be integrated for hostel fee payments.

Notifications

Email or SMS notifications could be added for:

- Complaints
- Maintenance updates
- Payment reminders
- Service request updates

Advanced Dashboard

Charts and statistics could be added to show:

- Hostel occupancy
- Available beds
- Pending complaints
- Maintenance requests
- Payment information

QR-Based Visitor Management

Visitors could be registered using QR codes for faster entry and exit management.

Cloud Deployment

The application could be deployed online so that authorized users can access it remotely.

PAGE 15 – CONCLUSION AND REFERENCES

10. CONCLUSION

The **Hostel Accommodation and Student Services Management System** successfully demonstrates the development of a database-driven hostel management application.

The system provides a centralized platform for storing and managing information related to students, hostels, rooms, beds, allocations, payments, complaints, maintenance requests, services, service requests, staff, visitors, and vacating records.

The use of **MySQL** allows the system to store structured information and maintain relationships between different hostel entities.

The **Python Flask** application provides the interface between the users and the database. The system also implements different roles for students, wardens, and administrators.

Students can access accommodation information and submit complaints, maintenance requests, and service requests. Wardens can manage accommodation and handle submitted requests. Administrators can manage important student and payment information.

The project demonstrates how database management concepts can be applied to a real-world hostel management scenario.

Overall, the system provides a simple and organized approach to hostel accommodation and student service management and can be further enhanced with online payments, notifications, mobile applications, analytics, and cloud deployment.

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